

UBUNTU

I AM BECAUSE WE ARE



IMEDIA PRODUCTION 2 - CS6042

Modular and Interactive Sculpture

Ubuntu Cubes - Interactive Sensor Installation exploring unity, collective effort, and the relationship between individual action and shared experience

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01 – INTRODUCTION & ARTIST STATEMENT

I am because we are.

Ubuntu Cube is an interactive installation in which four laser-cut cubes, with three embedded with an LED and a light-dependent resistor (LDR), are arranged on a 1×3 linear laser-cut board. The system is inspired by the African philosophical concept of Ubuntu, which means, '**I am because we are**', expressing the idea that individual elements only reach their full potential through collective effort and connection.

The project evolved from an earlier concept called TactiGrid, a 3 by 3 grid based installation in which cube placement detected positional alignment, inspired by the Tic Tac Toe game. Over the course of the development, the concept was simplified, refocused, and redesigned, moving from a game based interaction model to a philosophical, audiovisual one. The final installation connects physical interaction through electrical circuits to real time visual and sonic output in TouchDesigner, creating a seamless experience that rewards collective engagement.

“A person is a person through other person.”

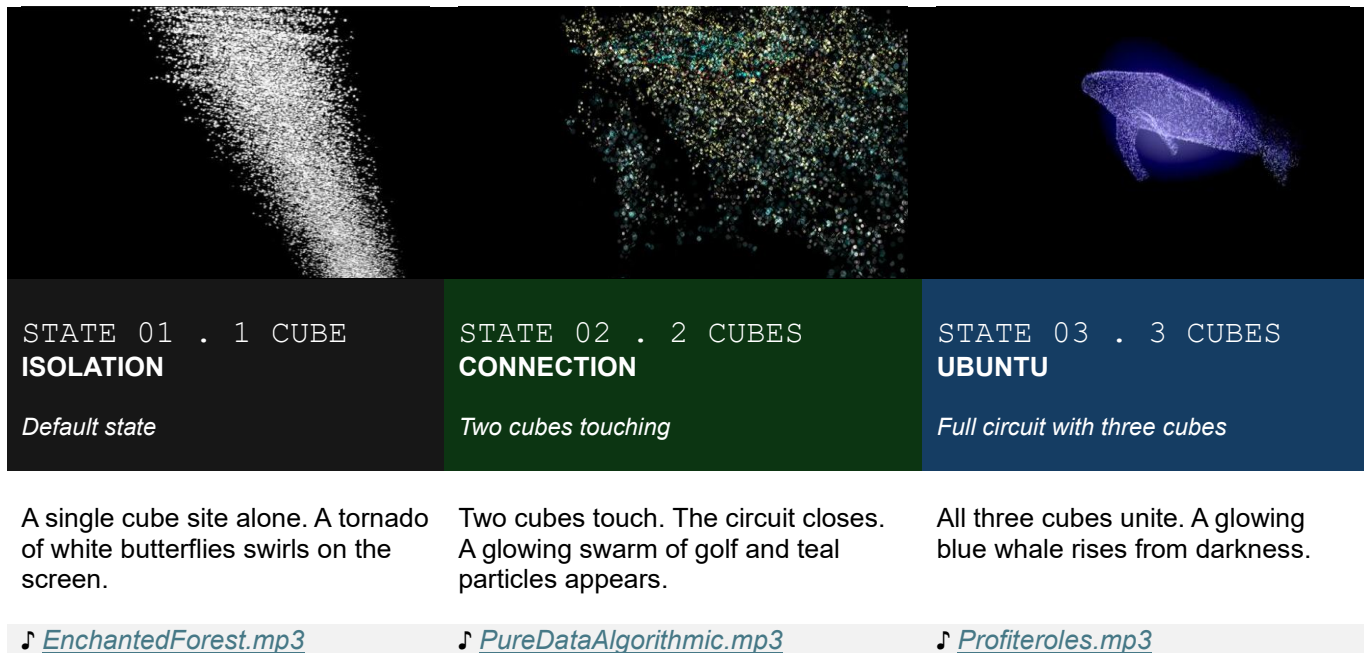
Archbishop Desmond Tutu on Ubuntu

The animals we chose represent states of human spirit. The butterfly tornado is the solitary self, beautiful and restless but also chaotic and disoriented. The particles swarm is two minds beginning to converge, finding shape and adding colours to each other. The whale is the fullest expression of Ubuntu in the natural world: whales travel in pods, communicate across oceans and grieve together. They are community made flesh.

02 – STORYBOARD & CONCEPT

Ubuntu made tangible.

The interactive rules of the sculpture follow a single principle: the more cubes that connect, the richer and more alive the experience becomes. This is Ubuntu made tangible, the audience discovers it by doing, not by reading instructions.



Design Principle

The audience is never told what to do. There are no instructions. The sculpture communicates through change; each new connection is its own reward, and the full experience is only possible together.

Concept Evolution – From TactiGrid...

TactiGrid was envisioned as a modular interactive installation composed of a 3 by 3 grid board with three cubes, an Arduino microcontroller, and a laptop responsible for sound generation. The user would interact with the system by placing cubes on the grid, and once the cubes are positioned, the system would detect each cube's location through copper contacts on both the cubes and the board, and respond in real time by activating light within the cubes and generating sound through the laptop. The initial concept drew inspiration from the familiar structure of Tic Tac Toe, using it as a spatial interface. The familiar grid was intended to make interaction immediately intuitive. The system translated position into audiovisual output, making the interaction visible and audible without any instructions or screen-based UI.

The original system proposed two distinct states based on cube alignment:

1. **Unaligned state:** When cubes are not placed in perfect alignment, each cube emits a different colour light. The sound output plays three separate dissonant, unpleasant sounds which express separation and disorder.
2. **Aligned state:** When the three cubes form a horizontal, vertical, or diagonal line on the grid, all cubes turn green simultaneously and the sound output plays a single unified harmonic sound, described as a unified expression of harmony and resolution.

There was no scoring system or end state. The interaction was intended to be open-ended and exploratory. The intended technical design specified copper coated cube faces connecting to copper contacts embedded in the 3 by 3 board cells. An Arduino would read these connections to determine cube positions,

evaluate whether an alignment condition was met, and send data to a processing environment that would handle sound generation. Each cube was to contain a single LED light controlled by the Arduino output pin.

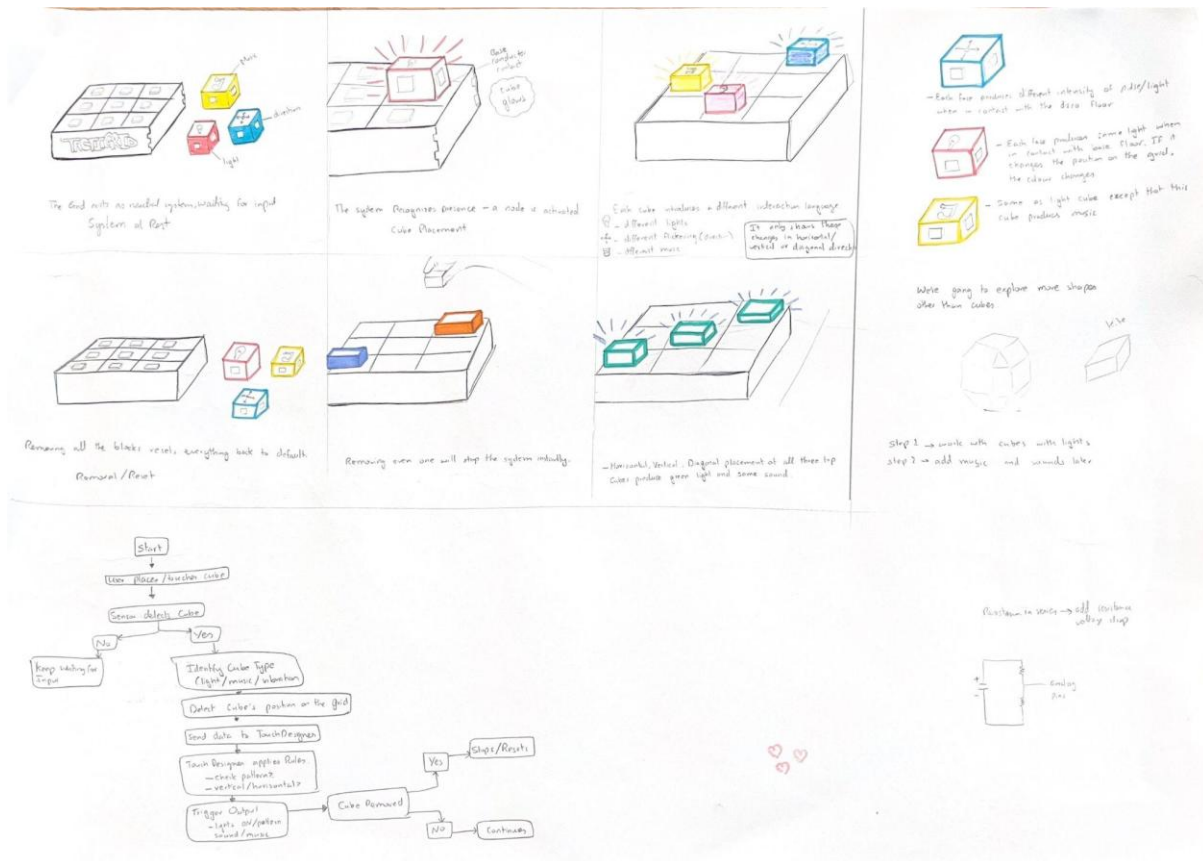


Figure 1: Storyboard and flow chart

...to Ubuntu

The project's theme was transformed from a game inspired concept (TactiGrid / Tic Tac Toe) to the African philosophical concept of Ubuntu. Derived from the Nguni Bantu phrase 'Umuntu ngumuntu ngabantu'; a person is a person through other persons. Ubuntu expresses the idea that individual identity, strength, and meaning are inseparable from community and collective relationship (Tutu, 1999).

This theme is particularly resonant in the context of interactive installation art, which has a rich tradition of using physical participation to evoke philosophical and social ideas (Rokeby, 1996). The installation does not instruct the user; rather, it invites them to discover the relationship between individual action and collective outcome through direct, embodied experience.

Our installation explored how human connection and community can influence individual emotional and social experiences. Rather than representing people as isolated entities, the project examined how relationships, proximity, and interaction between individuals can gradually create connection, communication, and shared presence. Within the installation, each cube represented a different individual with its own unique identity and "inner world." This was visually expressed through the illustrated dark spirit-animal characters placed on the surfaces of the cubes. Although every character appeared visually distinct,

strange, or chaotic, they collectively symbolised the complexity and uniqueness of human personalities and emotional states. The use of unsettling but expressive figures was intentional, aiming to communicate that every individual carries personal emotions, struggles, and experiences beneath the surface. Meanwhile, healing hand symbols represented the soothing influence of solidarity while also functioning as an instruction for the audience to interact with the installation.

The interaction between the cubes became the central metaphor of the project. When the cubes remained separated, no connection was formed between them. However, as the cubes physically moved closer together, conductive contact points allowed a circuit to form, causing lights inside the cubes to activate. This interaction symbolised the emergence of human connection, suggesting that relationships are formed through closeness, interaction, and mutual presence. The gradual “lighting up” of the installation acted as a visual representation of emotional and social connection.

Specifically, Ubuntu manifests as follows:

1. Each cube, in isolation, produces its own light and its own sound which signifies an individual voice.
2. When all three cubes are placed together on the board, their sensors interact, their signals combine, and the audiovisual output reflects a unified, collective state.
3. The transition from separate to unified translates from individual readings to shared experience which enacts the Ubuntu principle physically and sensorially.



Figure 2: Team with the Healing Hands making an Ubuntu sign (left) and the final version of Ubuntu (right)

03 – DESIGN CONSIDERATIONS & TECHNICAL ITERATIONS

Iteration and Refinement.

The path from TactiGrid to Ubuntu Cube involved a series of deliberate design decisions, technical explorations, and practical problem solving steps. Each iteration refined the system to fit time constraints, provide simplicity, and an expressive depth to the project.

Circuit Architecture – From Series to Parallel

The first circuit exploration proposed wiring each cube's internal components in a single series chain in the image below.

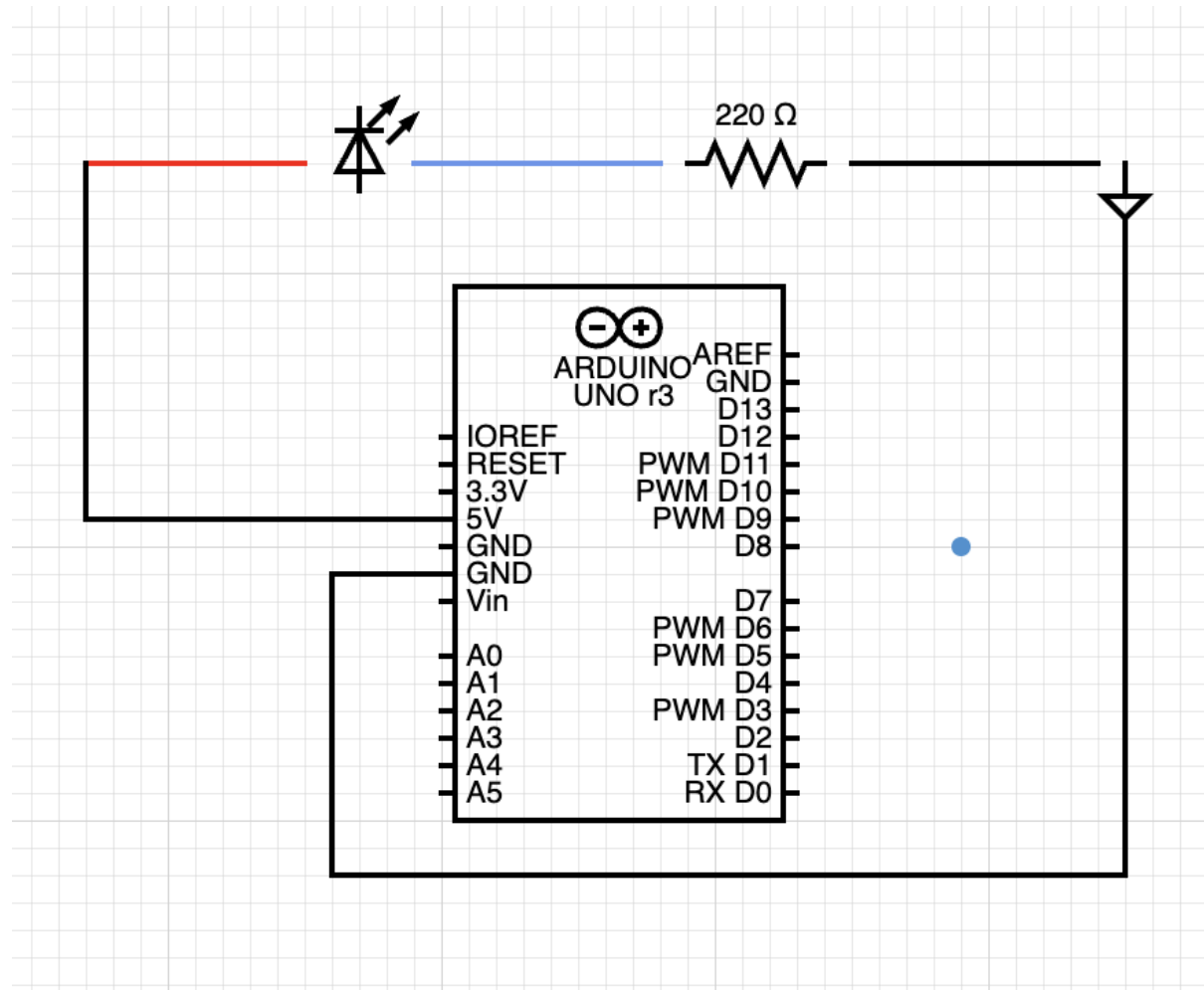


Figure 3: Arduino and LED circuit diagram

5V → LED → resistor → GND

While conceptually simple, this approach presented a significant voltage distribution and wiring problem. This design meant creating copper faces for each cube placement on the laser cut board, resulting in a huge network of wires, adding complexity to the project.

After analysis, the design was revised to use two independent parallel branches inside each cube consisting of the LED branch and LDR branch. Introducing LDRs to the project meant we could keep it simple and interactive by using light detecting sensors to create audio-visual output to correspond with user input.

1. The LED branch featured 5V → 220Ω resistor → LED → GND. This provides a constant, reliable current to the LED regardless of light conditions.

2. LDR branch: 5V → LDR → midpoint node → 10kΩ pull-down resistor → GND. The midpoint voltage, which varies with ambient light, is read by the Arduino's analog input (A0).

The internal wiring setup within each cube looked like the diagram below. The 5V line feeds power to the cube, while the LED and resistor branches maintain their own resistors, connecting back to GROUND. The midpoint line connects the LDR to the Arduino Analog pin which reads the data coming from the sensor.

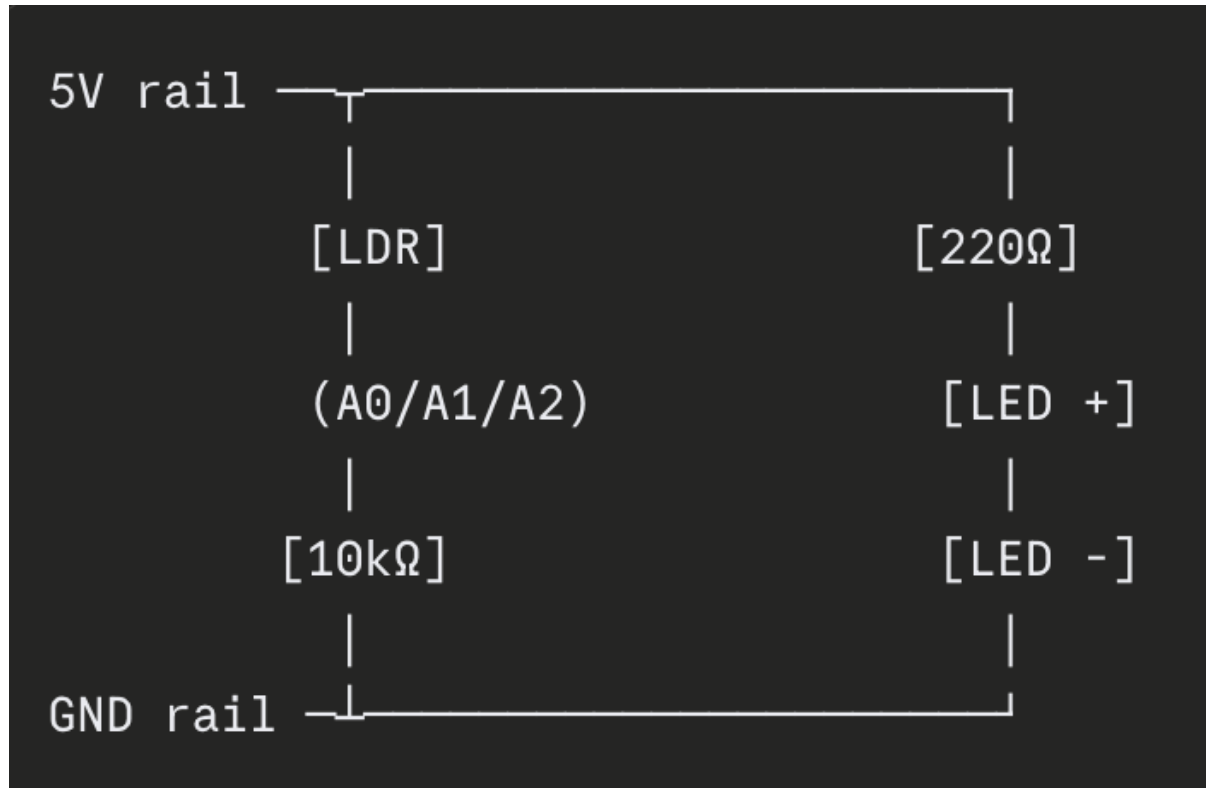


Figure 4: Internal wiring of the circuit

This separation of components keeping illumination and sensing independent, improved both reliability and design flexibility (Banzi & Shiloh, 2014).

LED Control

Redirecting from Pin 9 to 5V

Initially, the LED branch was designed to be switched by Arduino digital Pin 9, allowing software control of when the LEDs illuminate. This was reconsidered during implementation. Since the desired behaviour was for LEDs to remain on whenever the system is powered, and in order to simplify the circuit and reduce software dependency, the LED branch was connected directly to the Arduino's 5V pin. This eliminated the need for any digital output control code for illumination, making the circuit more robust and reducing the risk of software errors affecting the physical output.

Power Distribution

With three cubes needing power, we applied a daisy chain approach which was adopted for distributing 5V and GND through the copper face contacts between cubes. Two conductive contacts on each shared cube face with one carrying 5V and the other one carrying GND, allow the power rails to pass from cube to cube when the cubes are placed in line on the board. Each cube taps from these rails internally for both its LED and LDR branches. This approach required only two conductive contacts per connecting face, keeping the physical connector design minimal and reliable.

LDR Sensing

The original sensing design allocated one analog input pin per LDR (A0, A1, A2), allowing independent reading of each cube's light level. During iteration, the option of averaging all three LDR signals onto a single analog pin was explored. By connecting all three LDR midpoints together and routing them to A0, the Arduino reads a blended average of all three sensors simultaneously. While this sacrifices individual cube identification, it proved sufficient for detecting the collective state of whether all cubes are covered or not which also aligned with the intended Ubuntu theme of collective rather than individual expression.

The Arduino reads all three LDR values simultaneously via analogue pins A0, A1, and A2, and sends them as comma-separated serial data to the computer using the following code. Individual LDR readings are passed to TouchDesigner, allowing each cube to generate its own distinct audiovisual response, which was a key creative decision in the final system due to technical blockers.

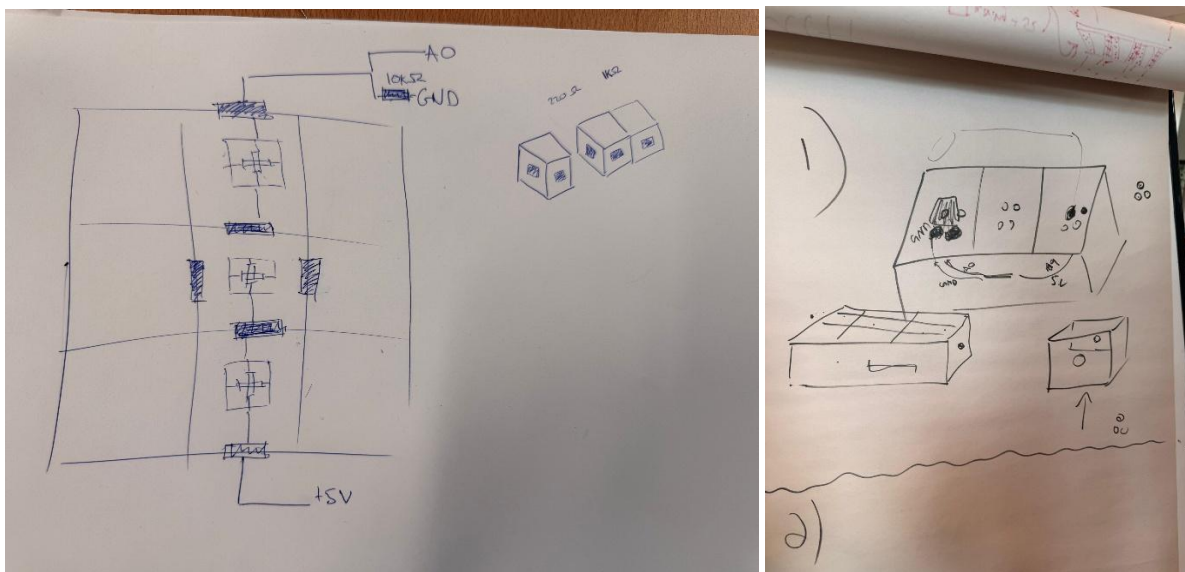


Figure 5: Sketches of the system

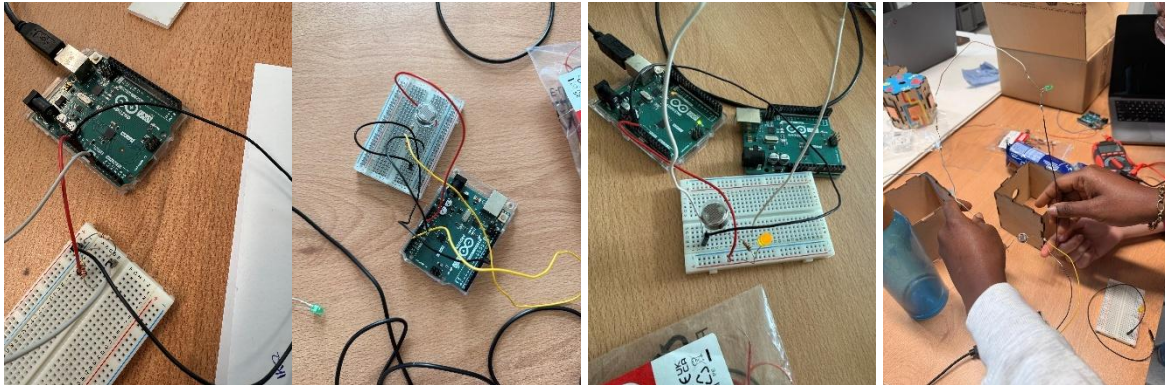


Figure 6: Circuit development

Circuit Design - Iteration 1

All LDR midpoints connected to separate analog pins on Arduino

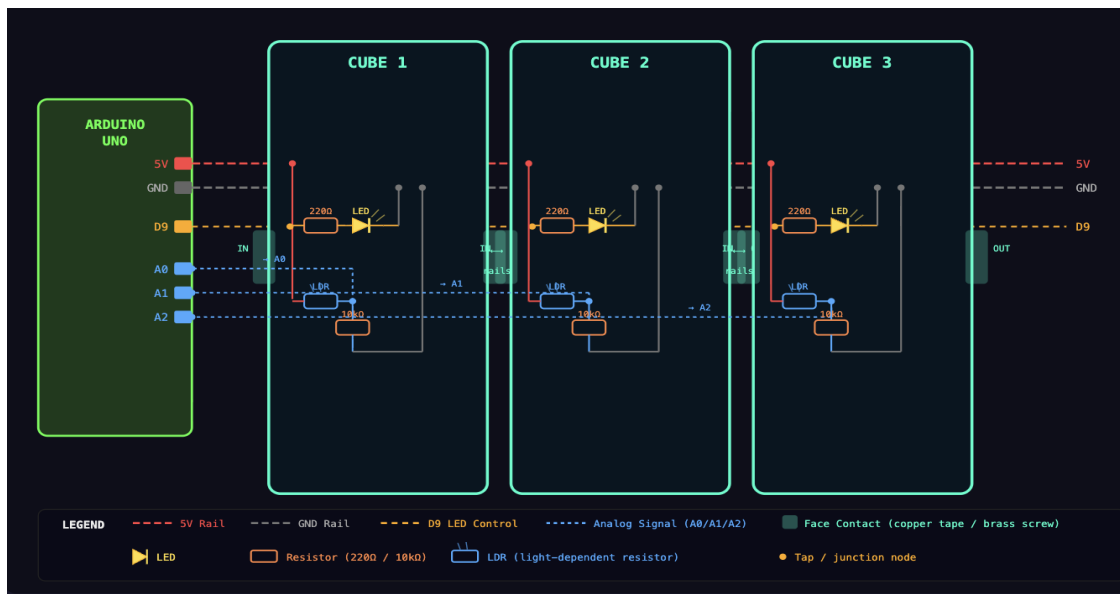


Figure 7: Circuit Iteration 1

Circuit Design - Iteration 2

All LDR midpoints connected to single analog pin on Arduino.

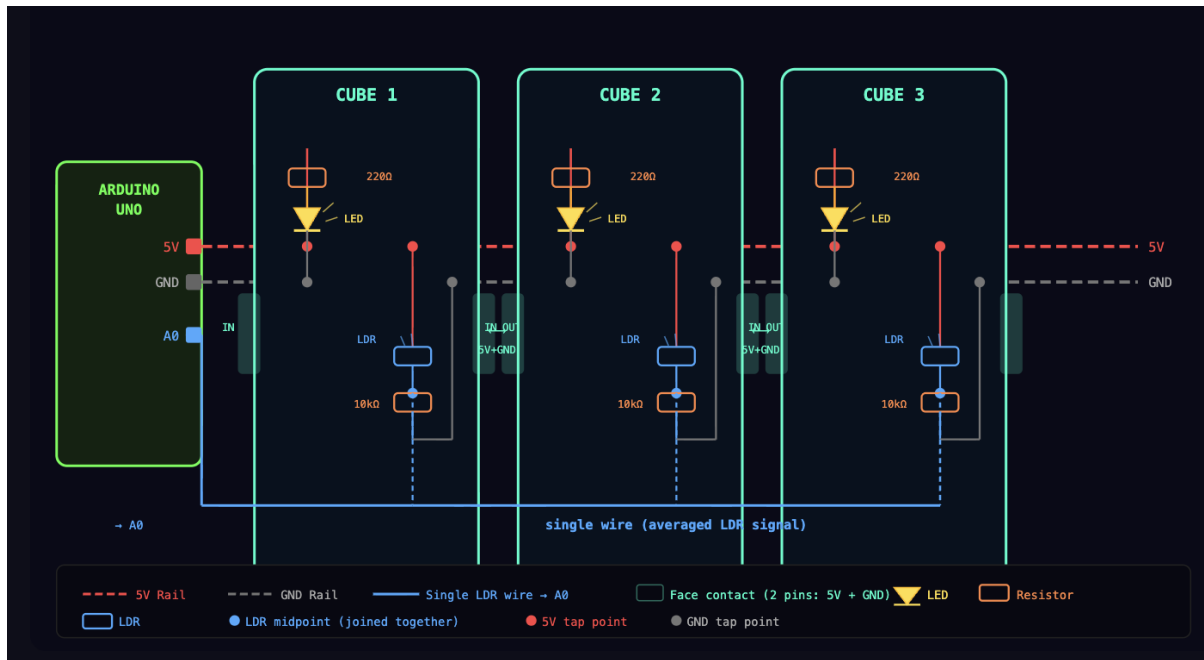


Figure 8: Circuit Iteration 2

Board Redesign (From Tic Tac Toe to Linear 1x3 board)

A significant structural iteration was the redesign of the board from a 3 by 3 grid to a simpler 1 by 3 linear arrangement. This change was motivated by several factors including the complexity of reliably detecting nine grid positions through copper contacts proved greater than anticipated within the project timeline. The linear arrangement aligned naturally with the concept of connection and unity given that three elements would form in one line. Also, the simpler board was more robust for a live installation context. The 1 by 3 board retains the essential spatial interaction, placing objects in a shared physical line, while reducing fabrication and wiring complexity.



Figure 9: Cubes being tested in a 3x3 grid and 1x3 grid

Building the Cuboids.

Technology Used

Hardware

Component	Role
Arduino Uno	Reads analog LDR values via A0; transmits data to TouchDesigner over USB Serial at 9600 baud
LED (5mm, green)	Provides illumination inside each cube; wired directly to 5V via 220Ω resistor
220Ω Resistor	Limits current through LED to safe operating level
LDR (Light Dependent Resistor)	Senses ambient light level inside each cube
10kΩ Resistor	Pull-down resistor forming voltage divider with LDR; midpoint voltage read by Arduino analog pin
Copper tape	Conductive face contacts on cube sides carrying 5V and GND rails between cubes on the board
Laser-cut MDF and acrylic covers	Physical housing for components. Features six faces with internal wire routing and copper contact pads
Laser-cut 1×3 linear board	Physical platform holding cubes in line; carries power rails and houses arduino and extra wiring

Table 1: Hardware components

Software

Component	Role
-----------	------

Arduino IDE	Writing, compiling and uploading firmware to the Arduino Uno; Serial Monitor for debugging
TouchDesigner (Derivative)	Receives serial data from Arduino and generates realtime visual and audio output with one channel assigned to each cube

Table 2: Software components

How the System Was Achieved

Physical Construction

Each cube was laser-cut from MDF and acrylic sheet. Two faces of each cube were fitted with copper tape contacts, which carried the 5V and GND rails. Internally, the LED and its 220 Ω resistor were soldered and mounted in the cube. The LDR was fitted in the cube and angled away from the LED to sense ambient light entering through the acrylic covers rather than the LED's own output. The 10k Ω pull-down resistor was soldered in series below the LDR midpoint to GND.

At first, the cube was made entirely from transparent acrylic plastic. However, we decided to change the material of the sides to MDF for better control over the light input via the sensor and more visible LEDs. This change also affected the appearance design of the cubes, as we decided to use black paint on MDF, allowing us to make the cubes more visually appealing with minimal impact on the system inside. Additionally, healing hand symbols were engraved in the base to further emphasise the Ubuntu theme. Cut-out patterns of the healing hand were also added to the cubes to make them more coherent.

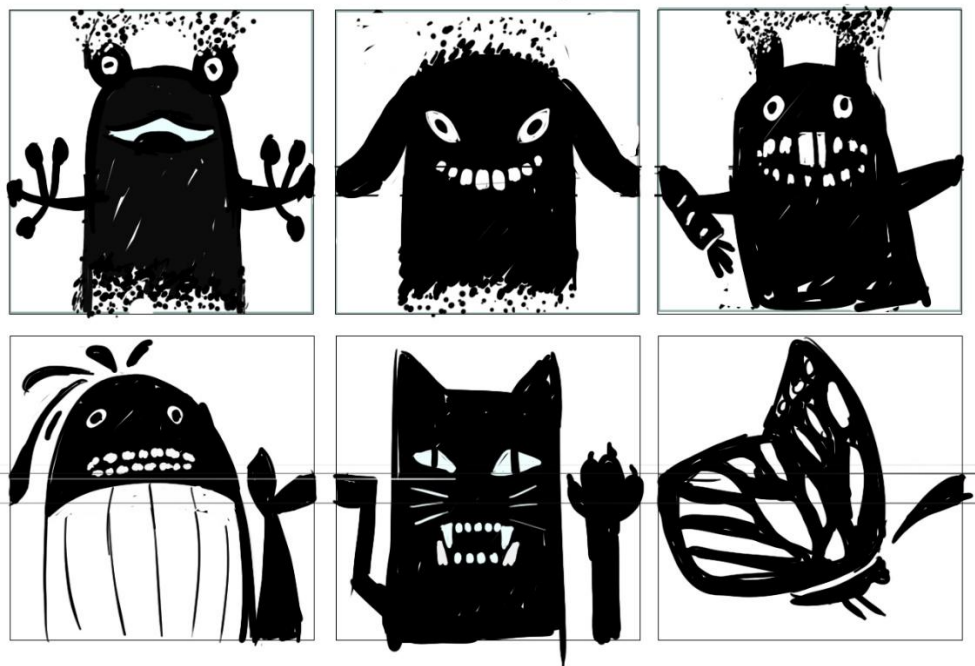


Figure 10: Digital sketch of the outer paintings

Copper contact points were embedded into the sides of the cubes, with wiring routed internally through openings designed into the cube walls. The copper plates were made with different shapes to easily differentiate between 5V and GND wires. The plates were designed to sit totally flat on the surface of the cubes, maintaining a stable connection and a neat overall appearance of the system. Furthermore, the wires inside the cube were also kept separate with a thin MDF piece for safety.

The outcome combined physical fabrication, electronics, symbolism, and interaction design into an installation that reflected loneliness, individuality, and the importance of human connection within communities. The project aimed to create an experience that was both visually expressive and conceptually reflective, encouraging viewers to think about how relationships shape personal identity and collective emotional experience.



Figure 11: The physical design of the cubes and the base



Figure 12: Shows the assembling process of the cubes

Assembling the Circuit

The three cubes were placed on the 1 by 3 board with their face contacts touching, completing the 5V and GND rails through all three. A single additional wire carried the averaged LDR signal from all three midpoint nodes, joined at a common point to the Arduino's A0 pin. The Arduino was powered via USB from the laptop running TouchDesigner, ensuring a shared ground reference across the entire system.

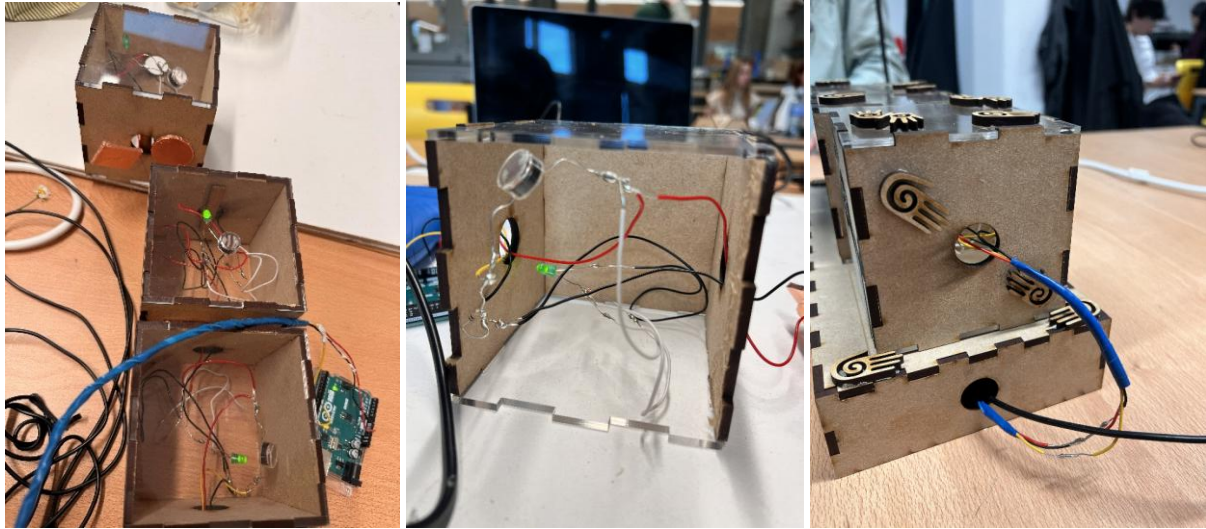


Figure 13: The cubes during the assembling phase

04 – TOUCHDESIGNER IMPLEMENTATION

The signal chain.

TouchDesigner is a node-based real-time visual programming environment. Every operation is an Operator (OP), and operators are wired together into chains. Data flows left to right through the network.

Before any animal visuals existed, the very first version of TouchDesigner patch used the LDR output to drive a simple expanding and contracting circle. Covering an LDR made the circle grow; uncovering it made it shrink. This was the proof of concept, confirming that the Arduino was sending clean data, the serial connection was stable, and the analog values were usable. Once this worked, we began with the whale, the particles, and the butterfly tornado on top of it.

Key Learning

The first few hours of the project were entirely about getting one sensor to move in one circle reliably. That boring foundation made everything else possible.

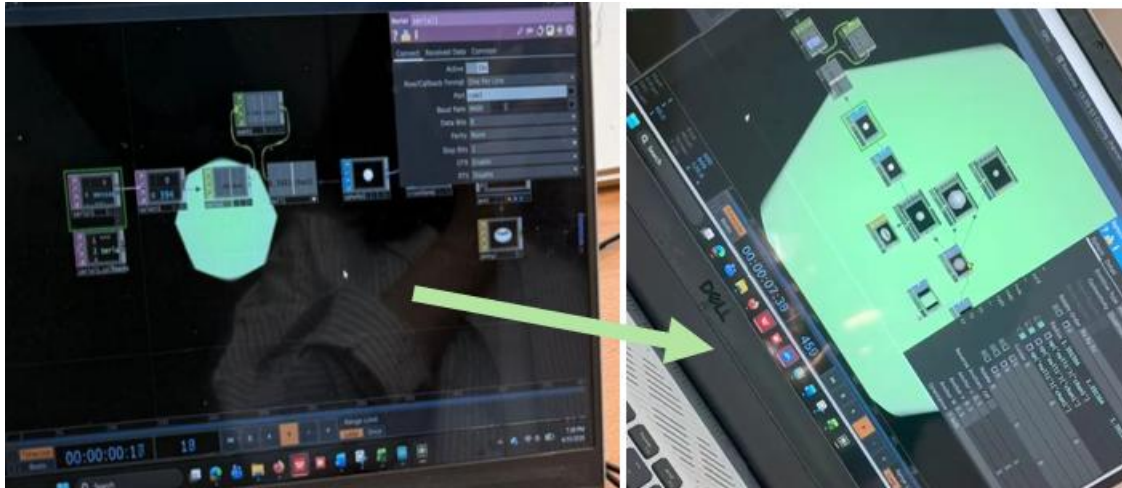


Figure 14: Shows the changes in the visuals in iteration 0 when the LDR would get different resistances.

Below is a complete breakdown of every key part of the network we built.

Step 1 – Receiving Arduino Data



Figure 15: Early TouchDesigner network, Serial DAT receiving Arduino data and then sending it forward to null chop

Node	Type	What it does
Serial DAT	DAT	Receives LDR values from Arduino over USB serial. The entry point for all physical data from the sculpture.
select1	CHOP	Picks one specific LDR value from the serial stream.
distol1	CHOP	Smooths the incoming signal to remove electrical noise so the visuals don't jump erratically.
math2	CHOP	Maps the raw Arduino values (0–1023) to a normalised 0–1 range using the Range tab. Everything downstream works in 0–1.

null4	CHOP	A Null CHOP — acts as a clean, stable branch point. Every part of the network that needs sensor data reads from here.
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Table 3: *Touch Designer components*

We used very simple code to connect the data received by the Arduino to TouchDesigner. When the resistance was high, the LED turned on, and when the resistance was low, the LED turned off.

```

1  int pirPin = 2;
2  int ledPin = 8;
3
4  void setup() {
5      pinMode(pirPin, INPUT);
6      pinMode(ledPin, OUTPUT);
7  }
8
9  void loop() {
10     int motion = digitalRead(pirPin);
11
12     if (motion == HIGH) {
13         digitalWrite(ledPin, HIGH); // LED ON
14     } else {
15         digitalWrite(ledPin, LOW); // LED OFF
16     }
17 }

```

Figure 16: Code to connect Arduino with TouchDesigner

Step 2 – The Whale Chain



Figure 17: The first part of the whale chain

The whale chain started with importing whale.fbx downloaded from [Quaternius](#) and the video was inspired by the work of Pablo Torri from his [YouTube tutorial](#).

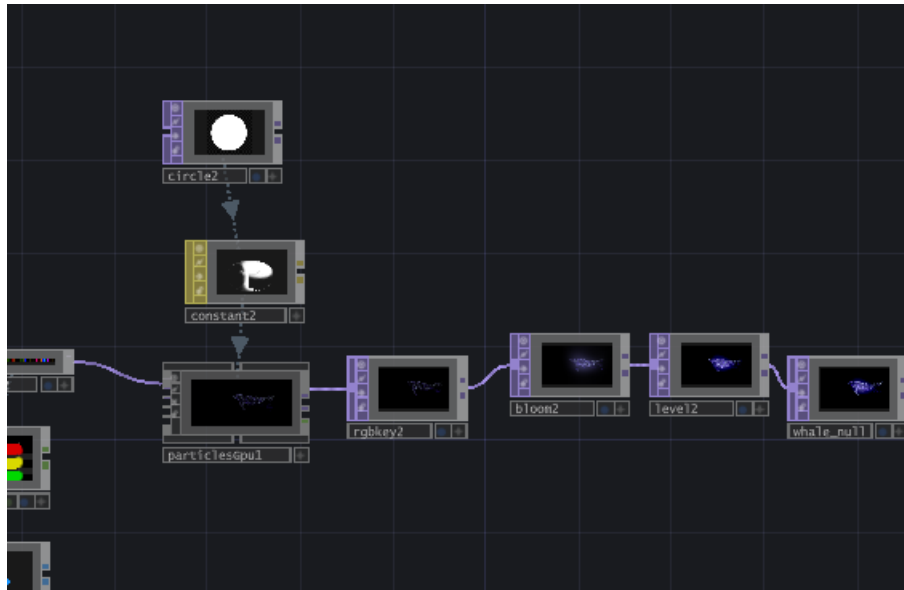


Figure 18: The second part of the whale chain

The whale's geometry scale parameter contains the Python expression `op('null2')[0]`. This reads the current value of null2 every frame, no physical wire is needed. When the LDR value rises (cube covered, darker inside), null2 rises, and the whale grows.

Node	Type	What it does
math1	CHOP	Scales the 0–1 sensor value from null4 into a useful size range for the whale geometry.
null2	CHOP	Reference null used specifically by the whale geometry. Expression <code>op('null2')[0]</code> drives the whale scale.
Whale SOP	SOP/TOP	The 3D whale model made of particles. Scale driven by <code>op('null2')[0]</code> . Renders to whale_null.
whale_null	TOP	Final rendered output of the whale chain. Feeds into Switch TOP as input 0.

Table 4: Touch Designer components for the whale effect

Step 3 – The Particles and the Butterfly chains

The butterfly chain works exactly like the whale chain except that the fbx file was a file of the butterfly and downloaded from [LasquetiSpice](#). We decided to play with the values and functions and ended up with a tornado like shape and that was exactly what we were looking for.

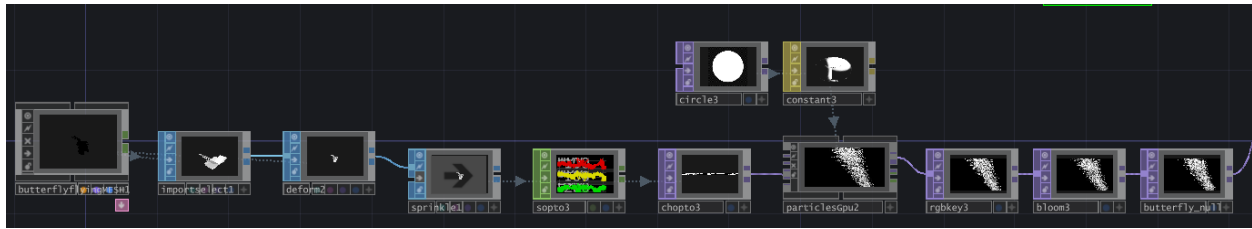


Figure 19: Shows the butterfly chain of nodes in TD

For the last chain, the particles chain, we just used ParticlesGPU and played with the values until we achieved the desired output.



Figure 20: Shows the Particles chain of nodes in TD

Step 4 – Counting Connected Cubes

This is the new logic that we added to make the visual switching work. The key challenge was converting three continuous analogue LDR values into a single integer count of how many cubes are covered.

To achieve that two things were done. The ranges and inputs were altered inside the math_threshold, math_count and null_count and then a code was used inside the Switch Top to change the index based on the values coming from all the three nulls, whale_null, particles_null and Butterfly_null.

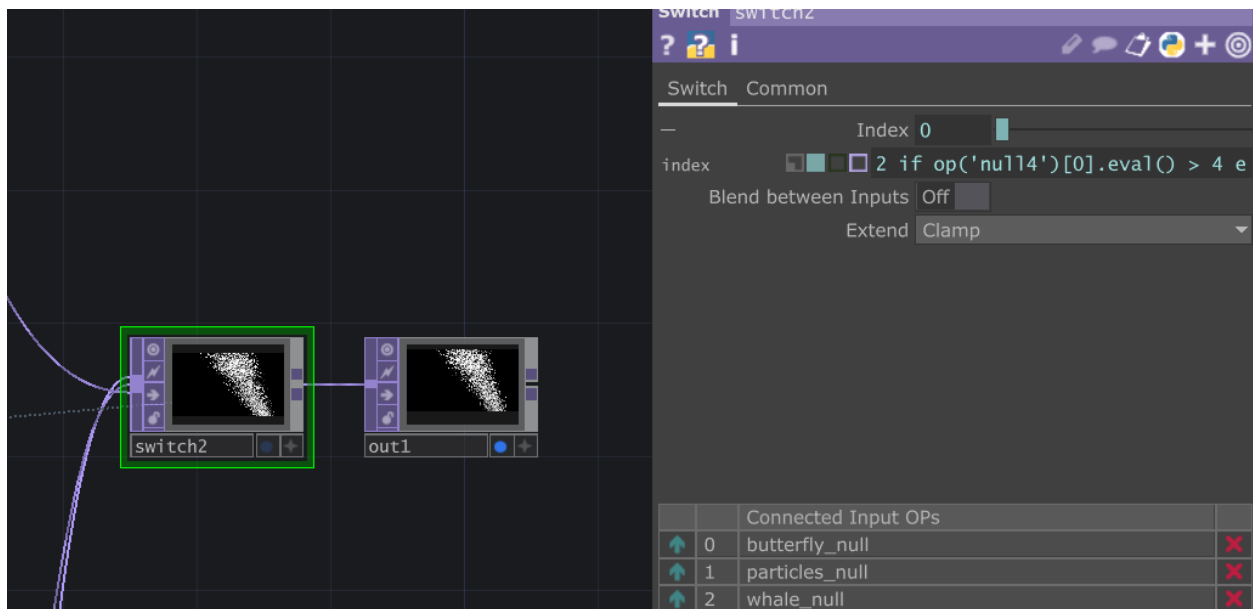


Figure 21: Shows the switch with three incoming nulls

The Switch TOP's index parameter is set to **2 if op('null4')[0].eval() > 4 else 1 if op('null4')[0].eval() > 3 else 0**. When null_count reads 2 (one cube), index 1 shows the whale. When it reads 1, index 1 shows particles. When it reads 0, index 2 shows the butterfly tornado. All three chains run continuously in parallel, the switch TOP simply selects which one the audience sees.

06 – AUDIO SYSTEM & PARALLEL VISUAL CHAINS

Three voices, One room.

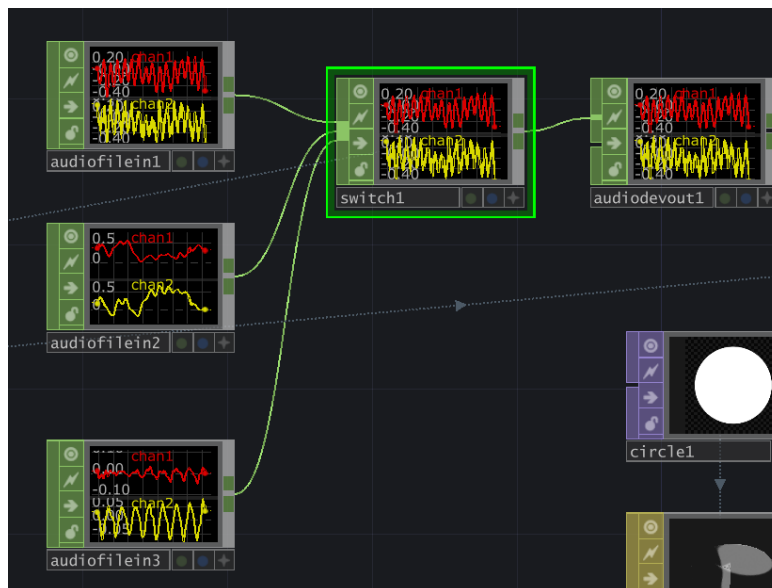


Figure 22: Shows the audio chain of nodes in TouchDesigner

Each visual state has its own dedicated track. The switching is done not with a physcia lselector but with Python expressions. Each audio File In CHOP has an audio file embedded. The Switch TOP's index parameter is set to **2 if op('null4')[0].eval() > 4 else 1 if op('null4')[0].eval() > 3 else 0**. When null_count reads 2 (one cube), audiofilein3 plays. When it reads 1, audiofilein2 plays. When it reads 0, audiofilein1 plays. Again, all three chains run continuously in parallel, the switch TOP simply selects which one the audience hears.

Node	Volume Expression	Active When
audiofilein1 (whale sound)	—	All 3 cubes connected
audiofilein2 (particles sound)	—	2 cubes connected
audiofilein3 (butterfly sound)	—	1 cube active
Mix CHOP / switch1	2 if op('null4')[0].eval() > 4 else 1 if op('null4')[0].eval() > 3 else 0	Combines all three; only one has volume at a time
audiodevout1	—	Sends mixed audio to speakers as output

Table 5: Touch Designer components for the audio

WHY A MIX CHOP?

The Audio Device Out CHOP accepts only two inputs. By routing all three Audi File In nodes through a Mix CHOP first, we bypass this limitation. Since only one audio file ever has volume above zero at a time, the output is always clean with no bleeding between tracks.

06 – THE THREE VISUALS

The Spirit Animals.

	State	Visual	Meaning	Sound
1 cube	Isolation	Butterfly tornado, white particles	The solitary self, fragile, searching	
2 cubes	Connection	Particle swarm, gold and teal	Two becoming one, form emerging from multiplicity	
3 cubes	Ubuntu	Glowing blue-white whale	Full community, ancient, vast, breathing	

Table 6: Touch Designer components for the visuals



Figure 23: State 01 - The butterfly tornado, default state, one cube active, white particles swarm



Figure 24: State 02 - The Particles swarm, two cubes connected, gold and teal particles finding form



Figure 25: State 03 - The Whale, all three cubes connected, glowing blue-white, size breathing with live sensor data

What we learned.

Reflections

The development of Ubuntu Cube demonstrates the iterative nature of physical computing projects, where conceptual ideas must be continuously tested against technical and material reality. The original TactiGrid proposal was rich in interaction potential but introduced complexity, particularly in detecting nine grid positions through copper contacts, which exceeded what could be achieved reliably within the project's constraints.

The decision to redesign the board from a 3 by 3 grid to a 1 by 3 linear arrangement helped with clarification. The linear arrangement more directly expresses the Ubuntu concept, which emphasizes three separate entities forming a single line of connection. The removal of the game-based alignment mechanic removed a layer of abstraction, making the interaction's meaning more comprehensible. The initial TactiGrid idea used a familiar game as a metaphor for interaction. However, Ubuntu uses a philosophical and cultural concept as the interaction itself. The user does not play a game rather, they enact a relationship through presence and collectivity.

The integration of TouchDesigner as the output medium proved expressive as its node-based visual programming environment allowed rapid iteration on the audiovisual response. The ability to assign distinct visual identities to each cube's sensor reading gave the installation an expressive granularity that a simpler output system would not have permitted.

Future Considerations

The current system provides a strong and functional foundation. Several directions for future development would extend its expressive and technical capabilities:

1. Expanded board configurations: The 1×3 board could be extended to a 1×5 or 2×3 arrangement, accommodating more cubes and enabling richer collective interactions and more complex Ubuntu narratives.
2. Wireless communication: Replacing the USB serial connection with Bluetooth or WiFi communication (using an ESP32 or HC-05 module) would remove the physical tether between the installation and the laptop, enabling more flexible spatial arrangements.
3. Dynamic LED colour control: Replacing fixed LEDs with RGB LEDs controlled via PWM pins would allow the cube's illumination colour to respond dynamically to sensor readings or system state, adding another layer of expressive feedback.
4. Multi-user interaction: Designing the board to accommodate more participants. For example, one cube per user in a group would scale the Ubuntu concept into a genuinely multi-person installation, where the collective state emerges from the actions of multiple individuals simultaneously.

The sculpture only makes sense together. So do we.

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09 – REFERENCES

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